Exercise Sheet 5

MATH 598 / 784 · High-Dimensional Probability

For discussion Friday, October 23, 2026

Throughout, $C, c > 0$ denote absolute constants, whose values may change. The sub-Gaussian parameter K uses the variance-proxy convention of the lectures.

1. Covariance estimation when the mean is unknown

Let $X \in \mathbb{R}^n$ have unknown mean μ and positive definite covariance Σ. Assume that, for every $v \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$,

$$\mathbb{E}e^{\lambda\langle X-\mu,v\rangle} \le \exp\!\left(\frac{K\lambda^2\langle \Sigma v, v\rangle}{2}\right), \qquad K \ge 1.$$

Given independent copies $X_1, \dots, X_m$, with $m \ge 2$, set

$$\bar{X} = \frac{1}{m}\sum_{i=1}^m X_i, \qquad \widehat{\Sigma}_m = \frac{1}{m}\sum_{i=1}^m (X_i - \bar{X})(X_i - \bar{X})^{\mathsf{T}}.$$

1. Prove the identity
$$\widehat{\Sigma}_m = \frac{1}{m}\sum_{i=1}^m (X_i - \mu)(X_i - \mu)^{\mathsf{T}} - (\bar{X} - \mu)(\bar{X} - \mu)^{\mathsf{T}}.$$
Compute $\mathbb{E}\widehat{\Sigma}_m$ and find an unbiased estimator of Σ.
2. Using a net of the unit sphere, show that, for every $u \ge 0$, with probability at least $1 - 2e^{-u}$,
$$\|\bar{X} - \mu\|_2^2 \le CK\|\Sigma\|_{\mathrm{op}}\frac{n+u}{m}.$$
3. Combine this with the covariance estimation theorem from Lecture 9 to prove that, with probability at least $1 - 4e^{-u}$,
$$\|\widehat{\Sigma}_m - \Sigma\|_{\mathrm{op}} \le CK\left(\sqrt{\frac{n+u}{m}} + \frac{n+u}{m}\right)\|\Sigma\|_{\mathrm{op}}.$$
Deduce a sufficient sample size for relative error $\varepsilon \in (0,1)$ with probability at least $1 - \rho$, where $\rho \in (0,1)$.

Hint. First bound the variance proxy of $\langle \bar{X} - \mu, v\rangle$. Recall that $\|zz^{\mathsf{T}}\|_{\mathrm{op}} = \|z\|_2^2$.

2. Missing directions and the coupon collector (Exercise 5.6.7)

Let $n \ge 2$, let J be uniform on $\{1, \dots, n\}$, and let σ be an independent uniform sign. Set $X = \sqrt{n}\,\sigma e_J$. For m independent copies, write $\Sigma_m = m^{-1}\sum_{i=1}^m X_i X_i^{\mathsf{T}}$.

1. Check that X is centered and isotropic and $\|X\|_2 = \sqrt{n}$ almost surely. Express Σ_m in terms of the numbers of samples in each coordinate direction.
2. Let Z count the coordinate directions that are never sampled. Show that
$$\mathbb{E}Z = n(1 - 1/n)^m, \qquad \mathrm{Var}(Z) \le \mathbb{E}Z, \qquad \mathbb{P}\{Z = 0\} \le \frac{1}{\mathbb{E}Z}.$$
3. Deduce that if $m \le \frac{1}{4}n\log n$, then
$$\mathbb{P}\{\|\Sigma_m - I_n\|_{\mathrm{op}} \ge 1\} \ge 1 - n^{-1/2}.$$
4. Explain why this does not contradict the covariance estimation theorem. More precisely, show that any K for which $\mathbb{E}e^{\lambda\langle X,v\rangle} \le e^{K\lambda^2\|v\|_2^2/2}$ for all λ, v must satisfy $K \ge cn/\log(2n)$.

Hint. The indicators of two different directions being missed have nonpositive covariance. Use $\log(1 - 1/n) \ge -2/n$ and examine $\langle X, e_1\rangle$.

For Problems 3 and 4, you may use Gaussian concentration in its moment generating function form: if $G \sim N(0, I_d)$ and $h : \mathbb{R}^d \to \mathbb{R}$ is L-Lipschitz, then

$$\mathbb{E}e^{\lambda(h(G)-\mathbb{E}h(G))} \le e^{\lambda^2 L^2/2} \qquad (\lambda \in \mathbb{R}).$$

3. The length of an anisotropic Gaussian vector

Let A be a nonzero real $k \times n$ matrix and $G \sim N(0, I_n)$. Write $a = \|A\|_{\mathrm{op}}$ and $b = \|A\|_F = (\sum_{i,j} A_{ij}^2)^{1/2}$.

1. Show that $g \mapsto \|Ag\|_2$ is a-Lipschitz and $\mathbb{E}\|AG\|_2^2 = b^2$. Deduce from the moment generating function bound that $\mathrm{Var}(\|AG\|_2) \le a^2$.
2. Prove that
$$0 \le b - \mathbb{E}\|AG\|_2 \le \frac{a^2}{b} \le a,$$
and hence, for every $t \ge 0$,
$$\mathbb{P}\{\big|\|AG\|_2 - b\big| \ge a + t\} \le 2e^{-t^2/(2a^2)}.$$
3. Define the *stable rank* $r(A) = b^2/a^2$. Show that, for every $\varepsilon > 0$,
$$\mathbb{P}\{\big|\|AG\|_2 - b\big| \ge \varepsilon b\} \le 2\exp\left(-\frac{\varepsilon^2 r(A)}{8}\right).$$
Compare $A = I_n$ with a rank-one matrix. What determines whether the length has small relative fluctuations?

Hint. Compare the coefficients of λ^2 at $\lambda = 0$. For part 2, factor $b^2 - (\mathbb{E}\|AG\|_2)^2$. In part 3, first consider $\varepsilon b \ge 2a$; otherwise the claimed bound is trivial.

4. Concentration on the cube (Exercises 5.2.11–12)

Let Φ be the standard normal distribution function, and define $T : \mathbb{R}^n \to [0,1]^n$ by

$$T(g) = (\Phi(g_1), \dots, \Phi(g_n)).$$

1. Show that $T(G)$ is uniform on $[0,1]^n$ when $G \sim N(0, I_n)$, and that T is $(2\pi)^{-1/2}$-Lipschitz for Euclidean distance.
2. Let U be uniform on $[0,1]^n$, and let $f : [0,1]^n \to \mathbb{R}$ be L-Lipschitz, where $L > 0$. Prove that
$$\mathbb{P}\{|f(U) - \mathbb{E}f(U)| \ge t\} \le 2\exp\left(-\frac{\pi t^2}{L^2}\right) \qquad (t \ge 0).$$
3. Let $B \subseteq [0,1]^n$ be measurable with volume at least $1/2$. Apply part 2 and its one-sided version to $f(x) = \mathrm{dist}(x, B)$ to show that, for every $t \ge 0$,
$$\mathbb{P}\left\{\mathrm{dist}(U, B) \ge \sqrt{\frac{\log 2}{\pi}} + t\right\} \le e^{-\pi t^2}.$$

Hint. First bound $\mathbb{E}\,\mathrm{dist}(U, B)$ using the lower tail and $\mathbb{P}\{\mathrm{dist}(U, B) = 0\} \ge 1/2$.

Problem 2 is adapted from Exercise 5.6.7 and Problem 4 from Exercises 5.2.11–5.2.12 of Roman Vershynin, *High-Dimensional Probability*, first edition. Problem 3 is the Gaussian case of Theorem 6.3.2, using the change of center in Exercise 5.2.4 with $p = 2$. The author provides the first-edition PDF free of charge. Exercise numbering differs in the second edition.